

Aman Sood

+91 62802 73367 | amansood2005@gmail.com | LinkedIn | GitHub

EDUCATION

Punjab Engineering College (PEC), Chandigarh

Aug 2023 – Jun 2027

B.E. in Electrical Engineering

Relevant coursework / focus: Data Structures & Algorithms, Object-Oriented Programming

Class 12 (CBSE): 85% | Class 10 (ICSE): 95%

PROJECTS

E32Log — Real-Time Vehicle Telemetry & OBD-II Logger | C++, Embedded Systems, Serial Protocols [GitHub](#)

- Designed and implemented a real-time telemetry pipeline that ingests and processes 10+ vehicle parameters over OBD-II on ESP32.
- Built continuous data capture for RPM, speed, coolant temperature, fuel efficiency, and related diagnostics with sub-second polling.
- Implemented wireless transmission and local logging supporting 1000+ live records for continuous monitoring and debugging.
- Applied OOP design (modular classes for acquisition, buffering, and output) and efficient data structures for low-latency diagnostics.

Intelligent Traffic Surveillance System | Python, OpenCV, YOLOv8, Edge AI, IoT

- Built an end-to-end real-time traffic monitoring system using YOLOv8 and DeepSORT for multi-vehicle detection and tracking.
- Implemented red-light violation detection with ROI/polygon-based counting and rule-based decision logic for alerts.
- Integrated Arduino-driven LED, buzzer, and LCD feedback for low-latency violation signaling at the edge.
- Added ambulance-aware exception handling to reduce false positives and improve fairness of automated enforcement.

E-Commerce Web Application | JavaScript, HTML, CSS, REST APIs

- Developed a responsive storefront with product listing, cart, and checkout flows using HTML, CSS, and JavaScript.
- Integrated frontend clients with REST APIs for catalog and cart operations, focusing on clean request/response handling.
- Improved page load and UI responsiveness through lean DOM updates and performance-minded frontend structure.

EXPERIENCE

Engineering Intern — Bharat Electronics Limited (BEL)

Jan 2026 – May 2026

- Analyzed operational datasets from substations, DG sets, and solar systems to surface performance issues and anomalies.
- Supported energy-data tracking and documentation workflows used for ISO 50001 compliance readiness.
- Contributed to sustainability measurement efforts recognized at the GreenTech Energy Awards.

TECHNICAL SKILLS

Languages: C++, Python, Java, JavaScript

Core CS: Data Structures & Algorithms (DSA), Object-Oriented Programming (OOP), Problem Solving

Backend & APIs: FastAPI, Node.js, .NET, REST APIs

Databases: PostgreSQL, MongoDB

Frontend: HTML, CSS, JavaScript

Tools & Other: OpenCV, Edge AI / IoT systems, Git, Linux basics